# Transit Equity and Built Environment Disparities: Omaha Metro vs. Council Bluffs

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# 1 Variables used in Council Bluffs Omaha comparison

- Transit Destination Accessibility (D5CR): Total number of jobs reachable within a 45 minute commute via public transit.
- Automobile Destination Accessibility (D5AR): Total regional jobs reachable within a 45-minute commute via personal automobile
- Percentage of Zero Vehicle households (Pct_ZeroCar): Percentage of households that own zero cars calculated from number of zero card households and the total number of households.
- Jobs per Household Diversity(D2A_JPHH): The ratio of jobs to households within a block group.
- Total Street Intersection Density (D3A): Total intersection per square miles.
- Jurisdiction: Binary classification derived from FIPS codes that represent either Pottawattamie county or Douglas and Sarpy counties.

# 2 Methods

Filtered for Omaha and Council bluffs. A standardized 12-digit Census Block Group Fips identifier (GEOID12) was constructed using state, county, census tract and block group so every neighborhood had a matching format identifier. The tigris package in R allowed the 12 digit id to be matched with spatial geometries from the Census bureau to allow for plotting.

## 2.1 Distribution Evaluation & Non-Parametric Significance

The distribution properties of our indicators was evaluated to check for normality. Distribution properties checked included mean,median, skew and kurtosis. As a result of all indicators featuring a high degree of skew and kurtosis a non parametric significance test (Mann-Whitney U) was chosen to compare Council Bluffs and Omaha. In order to control for type 1 error as a result of several comparisons a Bonferroni

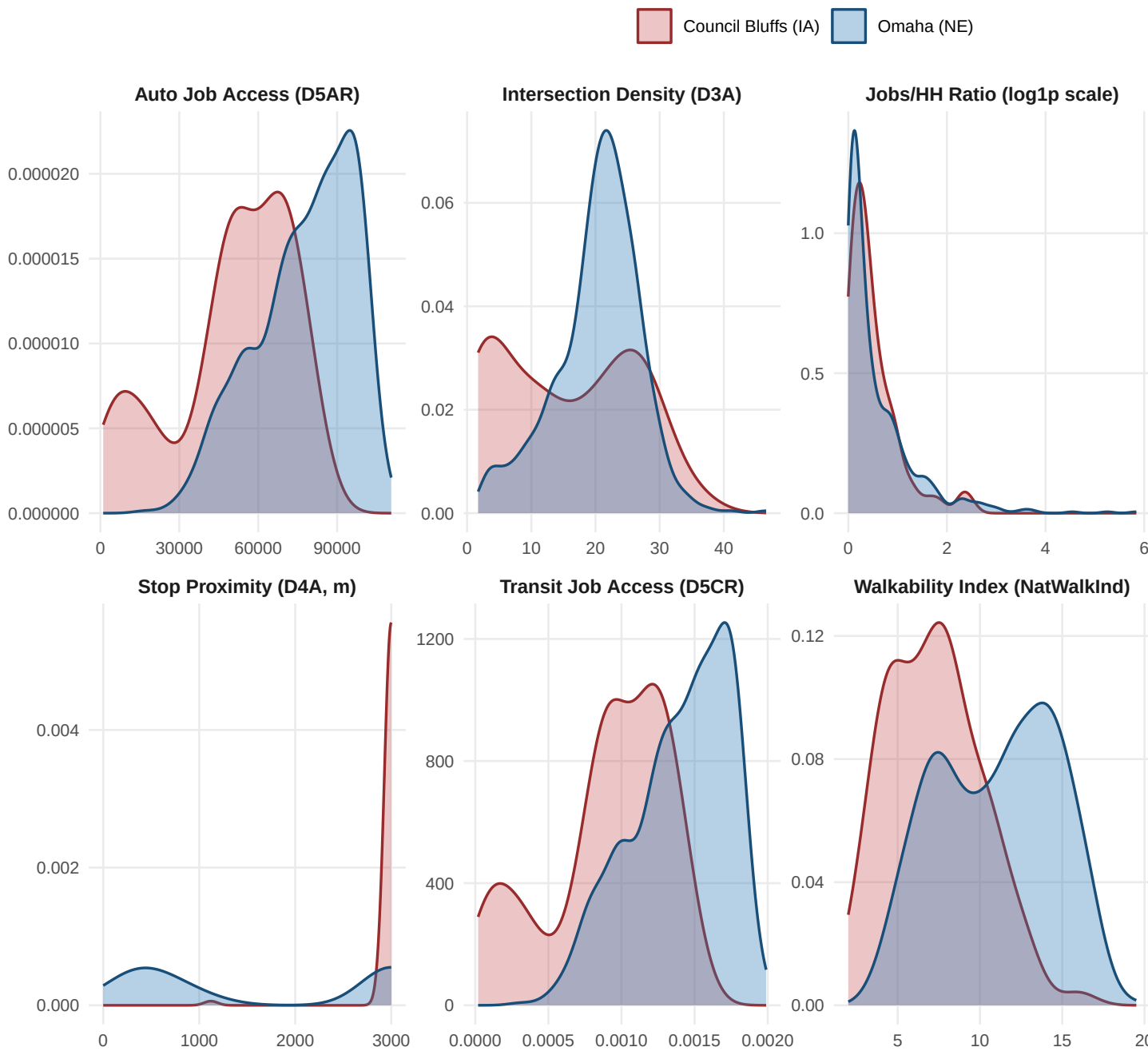
adjustment was applied. To determine practical significance an effect size was calculated using Cliff's delta (delta symbol).

Neighborhoods were grouped into low,medium and high for car-less households and bus-reachable jobs. The Y-axis was need for public transit in the form of how many people lived in a no car household. The X axis was how many jobs were reachable within a 45 minute drive meant to demonstrate how much transit was available to them. The Missouri river boundary was marked using the SF package to label the boundary between states.

# 3 Results:

## 3.1 Distribution and Normality

An evaluation of the eight transportation indicators were non- normal across both Omaha and Council bluffs (Table 1). This is a result of centrally located block groups having a larger share of transportation access and employment opportunists while those living outside these tracts had lower access. This result in a rejection of the assumption of normality. As a result a two tailed Mann-Whitney U test and cliffs delta effect size were chosen.




# 4 Figure 1: Distribution profiles of 8 indicators between both Council Bluffs and Omaha to display skeweness and normality of each variable.

## 4.1 Metropolitan Hypothesis tests

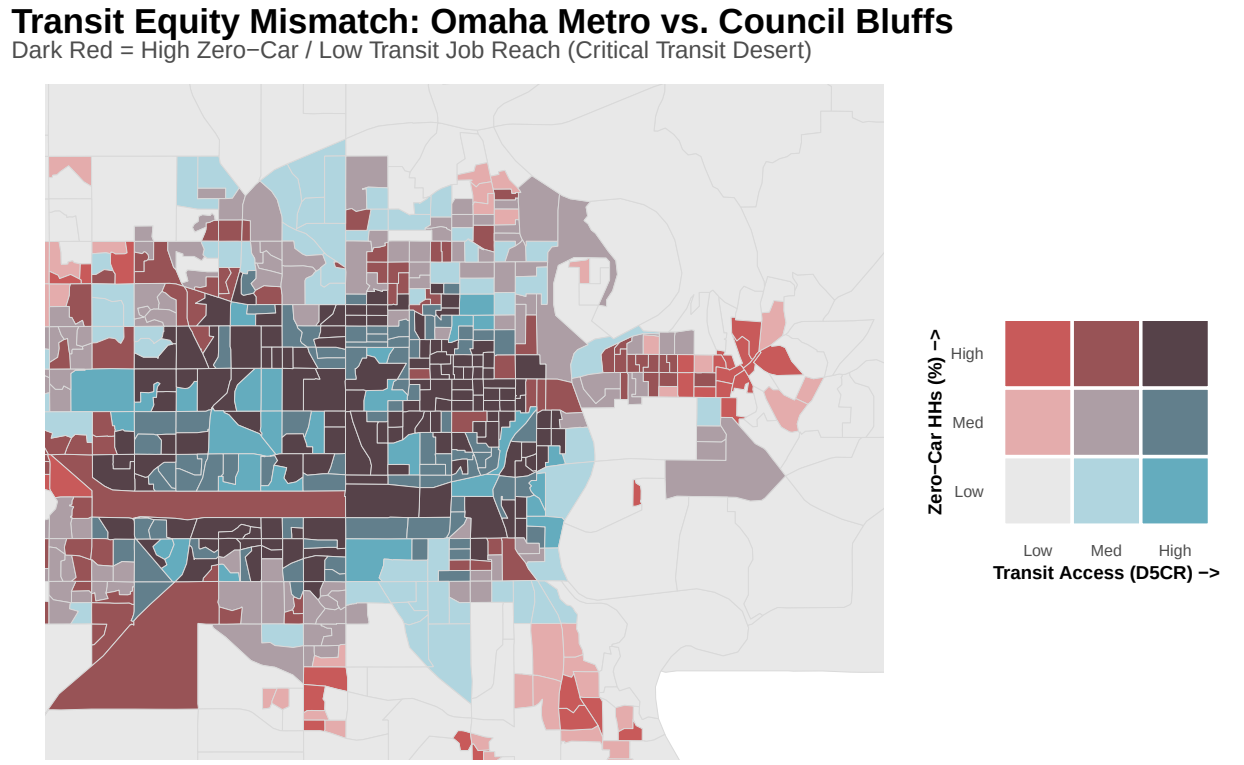
Hypothesis testing with a bonferroni correction identified several statistically significant differences between Omaha and Council Bluffs (Table 2). Omaha block groups were significantly better across 7 of the 8 indicators. Omaha block groups had significantly higher walk-ability, residential density, intersection density,

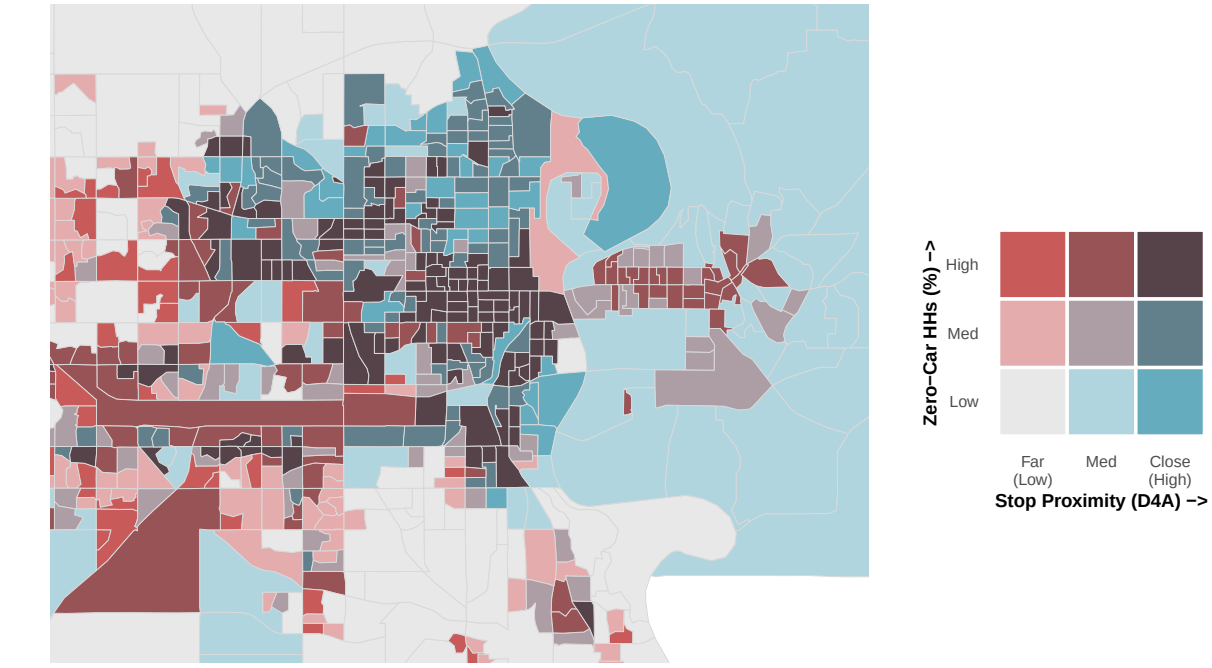
and transit stop proximity and percent of zero vehicle households. Omaha had significantly smaller distance to the nearest transit stop. There was no significant difference on jobs per household.

Table 1: Table 2: Non-Parametric Mann-Whitney U Tests and Cliff's Delta Effect Sizes (Bonferroni-Adjusted)

| Indicator | Dimension | Omaha Median | Council Bluffs Median | W Statistic | Bonferroni p-value | Sig. |
|---|---|---|---|---|---|---|
| NatWalkInd | Composite Walkability Index | 11.33 | 7.17 | 12661 | < 0.0001 | **** |
| D1A | D1: Residential Density (Units/Acre) | 2.75 | 1.66 | 20586 | < 0.0001 | **** |
| D2A_JPHH | D2: Diversity (Jobs/HH Ratio) | 0.33 | 0.39 | 30595 | 1 | ns |
| D3A | D3: Design (Intersection Density) | 21.13 | 14.59 | 21982 | 0.0006 | *** |
| D4A | D4: Transit Proximity (Stop Distance, m) | 873.74 | 3000.00 | 46122 | < 0.0001 | **** |
| D5AR | D5: Auto Dest. Accessibility (45m Jobs) | 81656.00 | 52881.00 | 10294 | < 0.0001 | **** |
| D5CR | D5: Transit Dest. Accessibility (45m Jobs) | 0.00 | 0.00 | 10294 | < 0.0001 | **** |
| Pct_ZeroCar | Auto: Zero-Vehicle Households (%) | 4.12 | 2.76 | 20045 | < 0.0001 | **** |

```
## Successfully matched block groups: 727
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##Bivariate transit deserts Consistent with the large effect size for transit access and job access decline sharply at the Nebraska Iowa border. Even though a high percentage of Council bluffs workforce commutes into Douglas and Sarpy county for work, council bluffs core has blocks with high amounts of 0 car households with poor access to jobs via transit as a result of limited transit options across the Nebraska Iowa border. While there exists some transit access from council bluffs into Omaha there exists need to expand transit options ro prevent some neighborhoods from being isolated from job access.

While transit job access is poor in council bluffs there is a high amount of proximity to physical transit stops. This coverage declines outside of downtown council bluffs with 40% of block groups exceeding 3,000 meters from a stop. For these residents walking to transit stops is not feasible.

Council bluffs suffers from significantly longer distances to transit stops, and for areas that do have transit stops they don't necessarily provide access to job opportunities. These factors contribute to council bluffs having a lower proportion of zero car households as car ownership is more important than the Omaha metro.

#Limitations The distance to transit stop variables was given an artificial cap of 3000 meters. While this helped to prevent extreme outliers it may have omitted the extent of isolation for those living further than 3000 meters from a stop. Similarly the 45 minute commute ceiling may not capture a broader trend of individuals in council bluffs doing a commute to work everyday that is greater than this ceiling.

## 4.2 Reproduction

To reproduce this analysis and generate all figures and statistical tables:

1. Ensure the raw EPA data file `EPA_SDL_OmahaCouncilBluffs.csv` is located in the working directory.
2. Open this `.Rmd` file in RStudio or an active R session[cite: 4].
3. Run the following command in the R console:

"'r rmarkdown::render("Final_council_bluffs_vs_omaha.Rmd")